Chapter 2. Simple Random Sampling

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1 Introduction

Definition 2.1

In **simple random sampling** (SRS) every sample of size n is equally likely; without replacement each of the $\binom{N}{n}$ samples has probability $1/\binom{N}{n}$.

A simple random sample can be drawn by the lottery method or with random number tables / computer programs (`sample(N, n)` in R).

Setting. The population values are $u_1, \dots, u_N$ with

$$\mu = \frac{1}{N}\sum_{i=1}^{N} u_i, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(u_i-\mu)^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(u_i-\mu)^2,$$

and $y_1, \dots, y_n$ is an SRS drawn without replacement (WOR). The **sampling fraction** is $f = n/N$ and $1-f$ is the **finite population correction** (fpc).

2 Estimation of the Population Mean

Motivation. The only randomness is which units are selected; the SRS mean is unbiased with finite population correction $(N-n)/N$.

Theorem 2.2: Sample mean under SRS (WOR)

Let $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2$. Then

(i) $\mathrm{E}(\bar{y}) = \mu$;

(ii) $\mathrm{Var}(\bar{y}) = \dfrac{N-n}{N-1}\dfrac{\sigma^2}{n} = \dfrac{N-n}{N}\dfrac{S^2}{n} = \dfrac{S^2}{n}(1-f)$;

(iii) $\mathrm{E}(s^2) = S^2$, hence $\widehat{\mathrm{Var}}(\bar{y}) = \dfrac{s^2}{n}(1-f)$ is unbiased for $\mathrm{Var}(\bar{y})$.

Derivation: proof of the theorem

Marginal distribution of one draw. By symmetry each y_i equals u_k with probability $1/N$, so

$$\mathrm{E}(y_i) = \sum_{k=1}^{N} u_k \frac{1}{N} = \mu, \qquad \mathrm{Var}(y_i) = \sum_{k=1}^{N} (u_k - \mu)^2 \frac{1}{N} = \sigma^2 \quad \text{for all } i.$$

(i) follows: $\mathrm{E}(\bar{y}) = \frac{1}{n}\sum \mathrm{E}(y_i) = \mu$.

Covariance of two draws. For $i \neq j$ the pair (y_i, y_j) is (u_k, u_l), $k \neq l$, with probability $\frac{1}{N(N-1)}$. Hence

$$\begin{aligned}
\mathrm{Cov}(y_i, y_j) &= \mathrm{E}(y_i y_j) - \mu^2 = \frac{1}{N(N-1)} \sum_{k \neq l} u_k u_l - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\
&= \frac{1}{N(N-1)}\Big[\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\
&= \frac{1}{N(N-1)}\Big[\frac{1}{N}\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] = -\frac{1}{N(N-1)}\Big(\sum_k u_k^2 - N\mu^2\Big) \\
&= -\frac{1}{N(N-1)} \sum_k (u_k - \mu)^2 = -\frac{\sigma^2}{N-1}.
\end{aligned}$$

Variance of the mean.

$$\begin{aligned}
\mathrm{Var}(\bar{y}) &= \frac{1}{n^2}\Big[\sum_i \mathrm{Var}(y_i) + 2\sum_{i<j} \mathrm{Cov}(y_i, y_j)\Big] = \frac{1}{n^2}\Big[n\sigma^2 - n(n-1)\frac{\sigma^2}{N-1}\Big] \\
&= \frac{\sigma^2}{n}\Big(1 - \frac{n-1}{N-1}\Big) = \frac{\sigma^2}{n}\frac{N-n}{N-1} = \frac{S^2}{n}\frac{N-n}{N},
\end{aligned}$$

using $N\sigma^2 = (N-1)S^2$.

Unbiasedness of s^2. Using $\sum (y_i - \bar{y})^2 = \sum (y_i - \mu)^2 - n(\bar{y} - \mu)^2$,

$$\mathrm{E}(s^2) = \frac{1}{n-1}\big[n\sigma^2 - n\mathrm{Var}(\bar{y})\big] = \frac{\sigma^2}{n-1}\Big(n - \frac{N-n}{N-1}\Big) = \frac{\sigma^2}{n-1} \cdot \frac{N(n-1)}{N-1} = \frac{N}{N-1}\sigma^2 = S^2.$$

Therefore $\mathrm{E}[\widehat{\mathrm{Var}}(\bar{y})] = \frac{N-n}{Nn}\mathrm{E}(s^2) = \mathrm{Var}(\bar{y})$.

Remark 2.3: SRS with replacement

If the draws are made with replacement, the y_i are iid with mean μ and variance σ^2. Then $\mathrm{Cov}(y_i, y_j) = 0$ (e.g. for 0/1 values $\mathrm{E}(y_i y_j) = P(y_i = 1)P(y_j = 1) = \frac{N_1}{N}\frac{N_1}{N}$), so

$$\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n}, \qquad \mathrm{E}(s^2) = \frac{1}{n-1}\Big(n\sigma^2 - n\frac{\sigma^2}{n}\Big) = \sigma^2, \qquad \widehat{\mathrm{Var}}(\bar{y}) = \frac{s^2}{n}.$$

Without replacement the variance is smaller by the factor $\frac{N-n}{N-1}$; the fpc matters only when f is not small.

Confidence interval and bound. A $100(1-\alpha)\%$ confidence interval for μ is

$$\bar{y} \pm z_{\alpha/2}\sqrt{1-f}\,\frac{s}{\sqrt{n}},$$

and with $\alpha = 0.05$ we use the 2-SD bound $B = 2\sqrt{\widehat{\mathrm{Var}}(\bar{y})} = 2\sqrt{\frac{s^2}{n}(1-f)}$.

Example 2.4: Tractors per village

Estimate the average number of tractors per village; $N = 69$, $n = 10$, $\bar{y} = 16.7$, $s^2 = 81.344$.

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{s^2}{n}\cdot\frac{N-n}{N} = \frac{81.344}{10}\cdot\frac{59}{69} = 6.956, \qquad 2\sqrt{6.956} = 5.27,$$

so the interval is $16.7 \pm 5.27 = [11.43,\ 21.97]$.

3 Estimation of the Population Total

Motivation. The total is N times the mean, with N^2 times its variance.

The population total is $\tau = N\mu$, and with N known the natural estimator is $\hat{\tau} = N\bar{y}$.

Theorem 2.5

$\mathrm{E}(\hat{\tau}) = \tau$ and

$$\mathrm{Var}(\hat{\tau}) = N^2\frac{N-n}{N-1}\frac{\sigma^2}{n} = N^2\frac{N-n}{N}\frac{S^2}{n}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2\frac{N-n}{N}\frac{s^2}{n} = N^2\frac{s^2}{n}(1-f).$$

A $100(1-\alpha)\%$ interval is $\hat{\tau} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$, typically $\hat{\tau} \pm 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$.

Example 2.6: Total number of tractors

$\hat{\tau} = 69 \times 16.7 = 1152.3$, $\widehat{\mathrm{Var}}(\hat{\tau}) = 69^2 \times 6.956 = 33{,}117$, and $2\sqrt{33117} = 364.0$. The interval is $1152.3 \pm 364.0 = [788,\ 1516]$.

4 Sample Size for Estimating μ and τ

To estimate μ with a bound B, set $2\sqrt{\mathrm{Var}(\bar{y})} = B$:

$$B = 2\sqrt{\frac{N-n}{N-1}\frac{\sigma^2}{n}} \implies n = \frac{N\sigma^2}{(N-1)D+\sigma^2} = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4}.$$

Derivation

Write $D = B^2/4$. Then $D\,n(N-1) = (N-n)\sigma^2$, i.e. $n[(N-1)D+\sigma^2] = N\sigma^2$. The second form follows from $\sigma^2 = \frac{N-1}{N}S^2$.

For the total, $B = 2\sqrt{N^2\frac{N-n}{N}\frac{S^2}{n}}$ gives

$$n = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4N^2}.$$

In practice S^2 (or σ^2) is unknown and replaced by a prior estimate or a pilot-sample s^2.

Example 2.7: Tractors, continued

With $s^2 = 81.344$ and $B = 3$ ($D = 2.25$),

$$n = \frac{69 \times 81.344}{69 \times 2.25 + 81.344} = \frac{5612.7}{236.6} = 23.72 \;\Rightarrow\; n = 24,$$

so 14 more villages are needed beyond the 10 already sampled.

5 Estimation of a Population Proportion

With $y_i = 1$ if unit i has the characteristic, 0 otherwise, $p = \mu$ and

$$\hat{p} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \hat{q} = 1 - \hat{p}.$$

For 0/1 values $\sigma^2 = pq$ and $S^2 = \frac{N}{N-1}pq$.

Theorem 2.8: Proportion under SRS

$$\mathrm{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{pq}{n}\frac{N-n}{N-1}, \qquad \widehat{\mathrm{Var}}(\hat{p}) = \frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N} = \frac{\hat{p}\hat{q}}{n-1}(1-f),$$

and $\mathrm{E}[\widehat{\mathrm{Var}}(\hat{p})] = \mathrm{Var}(\hat{p})$.

Derivation: why the divisor is $n-1$

$$\mathrm{E}(\hat{p}\hat{q}) = \mathrm{E}(\hat{p}) - \mathrm{E}(\hat{p}^2) = p - \left[\mathrm{Var}(\hat{p}) + p^2\right] = pq - \frac{pq}{n}\frac{N-n}{N-1}$$
$$= pq\,\frac{n(N-1) - (N-n)}{n(N-1)} = \frac{N(n-1)}{(N-1)n}\,pq.$$

Hence

$$\mathrm{E}\Big[\frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N}\Big] = \frac{N-n}{N(n-1)}\cdot\frac{N(n-1)}{(N-1)n}pq = \frac{pq}{n}\frac{N-n}{N-1} = \mathrm{Var}(\hat{p}).$$

By contrast the plug-in $\frac{\hat{p}\hat{q}}{n}(1-f)$ is biased downward: $\mathrm{E}\big[\frac{\hat{p}\hat{q}}{n}(1-f)\big] \neq \mathrm{Var}(\hat{p})$. This is the same phenomenon as s^2 versus the divisor-n variance, since $s^2 = \frac{n}{n-1}\hat{p}\hat{q}$ for 0/1 data.

Example 2.9: Student vote

An SRS of $n = 120$ from $N = 2400$ students; 72 support candidate A. Then $\hat{p} = 0.6$ and

$$\widehat{\mathrm{Var}}(\hat{p}) = \Big(1 - \frac{120}{2400}\Big)\frac{0.6 \times 0.4}{119} = 0.001916, \qquad 2\sqrt{0.001916} = 0.0875,$$

so the interval is $0.6 \pm 0.0875 = (0.512,\ 0.688)$.

6 Sample Size for Estimating p

Replace σ^2 by pq in the formula for μ:

$$n = \frac{Npq}{(N-1)D + pq}, \qquad D = \frac{B^2}{4} \quad (\text{or } D = B^2/1.96^2 \text{ with the exact normal quantile}).$$

With no prior information take $p = 1/2$, which maximizes pq and gives a conservative n.

Example 2.10

For $N = 2400$ and $B = 0.05$: $D = 0.05^2/4 = 0.000625$ and

$$n = \frac{2400 \times 0.25}{2399 \times 0.000625 + 0.25} = \frac{600}{1.7494} = 342.98 \;\Rightarrow\; n = 343.$$

7 Towards Stratified Sampling

Consider estimating the total first-hospitalization cost of kidney-stone patients. Two regions with very different incidence rates are surveyed separately:

	Carolinas	Rockies
n	363	258
$\bar{y}$	1350	1150

Pooling all 621 observations ignores the separate regional samples and population sizes; weight each regional mean by its population size (the stratified estimator, Chapter 3).

8 Selected exercises

Motivation. The SRS variance from inclusion indicators, the covariance of two sample means (used in Ch. 5), and a proportion computation.

Exercise 2.1 (Inclusion indicators)

Draw an SRS of size n without replacement from units $1, \dots, N$ with values $u_1, \dots, u_N$, and let $Z_k = 1$ if unit k is in the sample and $Z_k = 0$ otherwise.
(a) Show that $P(Z_k = 1) = n/N$ and, for $k \neq l$, $P(Z_k = Z_l = 1) = \frac{n(n-1)}{N(N-1)}$.
(b) Writing $\bar{y} = \frac{1}{n}\sum_{k=1}^{N} Z_k u_k$, rederive $\mathrm{E}(\bar{y}) = \mu$ and $\mathrm{Var}(\bar{y}) = \frac{S^2}{n}(1-f)$.

Solution. [Supplement]

(a) The samples containing k choose the other $n-1$ of $N-1$ units: $P(Z_k = 1) = \binom{N-1}{n-1}/\binom{N}{n} = n/N$; similarly $P(Z_k Z_l = 1) = \binom{N-2}{n-2}/\binom{N}{n} = \frac{n(n-1)}{N(N-1)}$.

(b)

Step 1. (Moments of the indicators.) By (a), $\mathrm{E}Z_k = f$, $\mathrm{Var}Z_k = f(1-f)$ (as $Z_k^2 = Z_k$), and for $k \neq l$

$$\mathrm{Cov}(Z_k, Z_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = \frac{n}{N}\Big(\frac{n-1}{N-1} - \frac{n}{N}\Big) = -\frac{f(1-f)}{N-1}.$$

Step 2. (Mean.) Only the Z_k are random, so $\mathrm{E}\bar{y} = \frac{1}{n}\sum_k f\,u_k = \frac{1}{N}\sum_k u_k = \mu$.

Step 3. (Variance.) Since $\sum_k Z_k = n$ is constant, $\bar{y} - \mu = \frac{1}{n}\sum_k Z_k(u_k - \mu)$; write $d_k = u_k - \mu$, so $\sum_k d_k = 0$ and $\sum_k d_k^2 = (N-1)S^2$. Then

$$n^2\mathrm{Var}(\bar{y}) = \sum_k d_k^2 \mathrm{Var}Z_k + \sum_{k\neq l} d_k d_l \mathrm{Cov}(Z_k, Z_l) = f(1-f)\Big[\sum_k d_k^2 - \frac{1}{N-1}\sum_{k\neq l} d_k d_l\Big].$$

Step 4. Because $\sum_{k\neq l} d_k d_l = \big(\sum_k d_k\big)^2 - \sum_k d_k^2 = -\sum_k d_k^2$, the bracket equals $\sum_k d_k^2\big(1 + \frac{1}{N-1}\big) = \frac{N}{N-1}(N-1)S^2 = NS^2$. Hence

$$\mathrm{Var}(\bar{y}) = \frac{f(1-f)NS^2}{n^2} = \frac{S^2}{n}(1-f).$$

The same computation with arbitrary inclusion probabilities π_k leads to the Horvitz–Thompson estimator $\sum_k Z_k u_k/\pi_k$ of τ. ■

Exercise 2.2 (Covariance of two sample means)

Each unit carries two values (x_k, y_k), $k = 1, \ldots, N$, with population means μ_x, μ_y and

$$S_{xy} = \frac{1}{N-1}\sum_{k=1}^{N}(x_k - \mu_x)(y_k - \mu_y).$$

For an SRS of n pairs without replacement, show that
(a) $\mathrm{Cov}(\bar{x}, \bar{y}) = \frac{1-f}{n} S_{xy}$;
(b) $s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$ satisfies $\mathrm{E}(s_{xy}) = S_{xy}$.

Solution. [Supplement] Apply the theorem on the sample mean under SRS to the auxiliary variable $z_k = x_k + y_k$, whose sample mean is $\bar{z} = \bar{x} + \bar{y}$.

(a)

Step 1. Expanding the square, $S_z^2 = \frac{1}{N-1}\sum_k[(x_k - \mu_x) + (y_k - \mu_y)]^2 = S_x^2 + S_y^2 + 2S_{xy}$.

Step 2. By the theorem, $\mathrm{Var}(\bar{z}) = \frac{1-f}{n} S_z^2$, $\mathrm{Var}(\bar{x}) = \frac{1-f}{n} S_x^2$ and $\mathrm{Var}(\bar{y}) = \frac{1-f}{n} S_y^2$.

Step 3. Since $\mathrm{Var}(\bar{z}) = \mathrm{Var}(\bar{x}) + \mathrm{Var}(\bar{y}) + 2\mathrm{Cov}(\bar{x}, \bar{y})$,

$$2\mathrm{Cov}(\bar{x}, \bar{y}) = \frac{1-f}{n}(S_z^2 - S_x^2 - S_y^2) = \frac{1-f}{n}\,2S_{xy}.$$

(b) The sample analogue of Step 1 is $s_z^2 = s_x^2 + s_y^2 + 2s_{xy}$. Taking expectations and using $\mathrm{E}(s_z^2) = S_z^2$, $\mathrm{E}(s_x^2) = S_x^2$, $\mathrm{E}(s_y^2) = S_y^2$ from the theorem,

$$2\mathrm{E}(s_{xy}) = S_z^2 - S_x^2 - S_y^2 = 2S_{xy}.$$

Polarization turns SRS variance statements into covariance statements (used for the ratio estimator bias, Ch. 5). ■

Exercise 2.3 (Hunters of game birds)

An SRS of $n = 1000$ of the $N = 99{,}000$ licensed hunters of a province is taken, and 430 of them hunt game birds.
(a) Estimate the proportion p of hunters seeking game birds and place a bound on the error of estimation.

(b) How large a sample is needed to estimate p with a bound $B = 0.02$, using the estimate of (a) as prior value?

Solution.

(a) $\hat{p} = 430/1000 = 0.43$ and, by the proportion theorem,

$$\widehat{\mathrm{Var}}(\hat{p}) = \frac{\hat{p}\hat{q}}{n-1} \cdot \frac{N-n}{N} = \frac{0.43 \times 0.57}{999} \cdot \frac{98{,}000}{99{,}000} = 0.000243, \qquad 2\sqrt{0.000243} = 0.031,$$

so p is estimated by 0.43 ± 0.031.

(b) With $D = B^2/4 = 0.0001$ and $pq = 0.43 \times 0.57 = 0.2451$,

$$n = \frac{Npq}{(N-1)D + pq} = \frac{99{,}000 \times 0.2451}{98{,}999 \times 0.0001 + 0.2451} = \frac{24{,}265}{10.145} = 2391.8,$$

so $n = 2392$ hunters are needed. Since N is large, this is close to the infinite-population value $pq/D = 2451$; the fpc saves about 60 interviews. ■